Lesson 4: Function Sequences and Metric Spaces

AI/ML Mathematics

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1 Motivation and Overview

Consider a reinforcement learning agent that iteratively improves its value function. Starting from an arbitrary bounded function V_0, the agent computes $V_1 = TV_0$, $V_2 = TV_1$, and so on, where T is the Bellman optimality operator. Three questions immediately arise: (1) Does the sequence of functions $\{V_n\}$ converge? (2) If so, what kind of convergence do we get—pointwise for each state, or something stronger? (3) Is the limit guaranteed to be the unique fixed point of T? These questions cannot be answered with sequences of real numbers alone; we need a theory of *sequences of functions* and, more fundamentally, a notion of distance between functions. The framework of *metric spaces* provides exactly this: by equipping the set of bounded functions with the supremum metric, we can treat value functions as points in a complete metric space and apply the *Banach fixed-point theorem* to establish both existence, uniqueness, and a geometric convergence rate. The tools developed in this lesson will let us prove that value iteration converges, quantify its rate, and understand precisely when limits commute with integration and other operations.

Why RL/ML needs function sequences and metric spaces

- **Value iteration.** The Bellman operator T is a γ-contraction on the complete metric space of bounded value functions under the sup-norm. The Banach fixed-point theorem guarantees convergence of $V_n = T^n V_0$ to the unique optimal value function V^* at rate $O(\gamma^n)$.
- **Policy evaluation.** Temporal difference methods generate sequences of value function estimates; uniform convergence ensures that continuity and integrability of the limit are preserved.
- **Fitted value iteration.** When value functions are approximated by neural networks, one works in function spaces equipped with metrics (e.g., L^2 or sup-norm), and completeness of these spaces is essential for convergence guarantees.
- **Monte Carlo integration.** The interchange of limit and integral, justified by uniform convergence, underpins the validity of approximating expected rewards by sample averages of convergent function sequences.

Prerequisites. Convergence of real sequences, the Cauchy criterion, supremum and infimum (Lesson 1, Definitions 3.4–3.5 and Theorem 4.10). Continuity (ε-δ characterization), uniform continuity, and the extreme value theorem (Lesson 2, Theorems 5.2 and 7.4). The Riemann integral, its properties, and the fundamental theorem of calculus (Lesson 3, Theorems 7.6–9.2).

Learning objectives. After completing this lesson, you will be able to:

- **Define** pointwise and uniform convergence of function sequences, and **explain** the key difference (dependence of N on x).
- **Prove** that the uniform limit of continuous functions is continuous and that uniform convergence permits interchange of limit and Riemann integral.
- **Apply** the Weierstrass M-test to determine uniform convergence of series of functions.
- **Define** metric spaces, open and closed sets, Cauchy sequences, completeness, and sequential compactness.
- **State** and **prove** the Banach fixed-point theorem, including existence, uniqueness, and the convergence rate estimate.
- **Justify** why value iteration converges by identifying the Bellman operator as a contraction on a complete metric space.

2 References

- Rudin [1976, Chapter 7] — Rigorous treatment of sequences and series of functions, uniform convergence, equicontinuity, and the Stone–Weierstrass theorem.
- Abbott [2015, Chapter 6] — Accessible development of pointwise vs. uniform convergence with many examples and the interchange theorems.
- Rudin [1976, Chapter 2] — Metric space topology: open and closed sets, compactness, completeness, and the contraction mapping principle.
- Bartle and Sherbert [2011, Chapters 8–9] — Detailed treatment of function sequences, metric spaces, and the contraction mapping theorem with applications.
- Pugh [2015, Chapter 2] — Metric spaces with a geometric viewpoint, completeness, and the Banach fixed-point theorem.

3 Pointwise and Uniform Convergence

We begin with the two fundamental modes of convergence for sequences of functions. The distinction between them—seemingly a minor quantifier swap—has profound consequences for the regularity of the limit.

Definition 3.1 (Pointwise Convergence)**.** Let $A \subseteq \mathbb{R}$ and let $\{f_n\}$ be a sequence of functions $f_n \colon A \to \mathbb{R}$. We say $\{f_n\}$ *converges pointwise* to $f \colon A \to \mathbb{R}$ if for every $x \in A$ and every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ (which may depend on *both* x and ε) such that

$$n \geq N \implies |f_n(x) - f(x)| < \varepsilon.$$

We write $f_n \to f$ pointwise on A.

Definition 3.2 (Uniform Convergence)**.** Let $A \subseteq \mathbb{R}$ and let $\{f_n\}$ be a sequence of functions $f_n \colon A \to \mathbb{R}$. We say $\{f_n\}$ *converges uniformly* to $f \colon A \to \mathbb{R}$ if for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ (depending *only* on ε) such that

$$n \geq N \implies |f_n(x) - f(x)| < \varepsilon \quad \text{for all } x \in A.$$

Equivalently, $\sup_{x \in A} |f_n(x) - f(x)| \to 0$ as $n \to \infty$. We write $f_n \rightrightarrows f$ on A.

Remark 3.3. Uniform convergence implies pointwise convergence (take the same N for every x), but the converse is false, as the next example shows.

Example 3.4 (Pointwise but Not Uniform Convergence)**.** Define $f_n \colon [0,1] \to \mathbb{R}$ by $f_n(x) = x^n$. For each fixed $x \in [0,1)$, $x^n \to 0$ as $n \to \infty$, and $f_n(1) = 1$ for all n. Thus $f_n \to f$ pointwise, where

$$f(x) = \begin{cases} 0 & \text{if } 0 \leq x < 1, \\ 1 & \text{if } x = 1. \end{cases}$$

Each f_n is continuous on $[0,1]$, but the limit f is *discontinuous* at $x = 1$. We claim the convergence is not uniform. Indeed, for any n, choosing $x_n = (1/2)^{1/n} \in [0,1)$ gives $f_n(x_n) = 1/2$, while $f(x_n) = 0$, so

$$\sup_{x \in [0,1]} |f_n(x) - f(x)| \geq |f_n(x_n) - f(x_n)| = \tfrac{1}{2} \not\to 0.$$

This example shows that pointwise convergence can destroy continuity. Uniform convergence prevents this, as we prove next.

Example 3.5 (Uniform Convergence on a Restricted Domain)**.** Consider the same functions $f_n(x) = x^n$, but now on $[0, a]$ for a fixed $a \in (0, 1)$. The pointwise limit is $f \equiv 0$, and

$$\sup_{x\in[0,a]} |f_n(x) - 0| = a^n \to 0$$

as $n \to \infty$. Hence $f_n \rightrightarrows 0$ on $[0, a]$. The limit is continuous, consistent with the theorem below.

Uniform Limit of Continuous Functions

Theorem 3.6 (Continuity of the Uniform Limit)**.** *Let $\{f_n\}$ be a sequence of continuous functions $f_n \colon A \to \mathbb{R}$ that converges uniformly to f on A. Then f is continuous on A.*

Proof. Fix $c \in A$ and let $\varepsilon > 0$. We need to find $\delta > 0$ such that $|x-c| < \delta$ implies $|f(x)-f(c)| < \varepsilon$. We split the difference into three parts:

$$|f(x) - f(c)| \le |f(x) - f_n(x)| + |f_n(x) - f_n(c)| + |f_n(c) - f(c)|.$$

Step 1. Since $f_n \rightrightarrows f$, choose N such that $\sup_{t\in A} |f_n(t) - f(t)| < \varepsilon/3$ for all $n \ge N$. Fix any such n. Then $|f(x) - f_n(x)| < \varepsilon/3$ and $|f_n(c) - f(c)| < \varepsilon/3$.

Step 2. Since f_n (for this fixed n) is continuous at c, choose $\delta > 0$ such that $|x - c| < \delta$ implies $|f_n(x) - f_n(c)| < \varepsilon/3$.

Combining:

$$|f(x) - f(c)| < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon.$$

□

Pointwise convergence does not preserve continuity

Example 3.4 shows that pointwise convergence of continuous functions can produce a discontinuous limit. The culprit is that the convergence rate $N(x, \varepsilon)$ may blow up as x approaches certain points (in that example, $x \to 1^-$). Uniform convergence eliminates this by requiring a single $N(\varepsilon)$ that works simultaneously for all x. This is the function-sequence analogue of the distinction between continuity and uniform continuity from Lesson 2.

The next theorem shows that uniform convergence also permits interchanging limits and Riemann integrals.

Theorem 3.7 (Interchange of Limit and Integral)**.** *Let $\{f_n\}$ be a sequence of Riemann integrable functions on $[a, b]$ that converges uniformly to f on $[a, b]$. Then f is Riemann integrable on $[a, b]$ and*

$$\lim_{n\to\infty} \int_a^b f_n(x)\,dx = \int_a^b f(x)\,dx.$$

Proof. Step 1: f is Riemann integrable. Given $\varepsilon > 0$, choose N such that $\sup_{x\in[a,b]} |f_n(x) - f(x)| < \varepsilon/(3(b - a))$ for all $n \ge N$. Fix such an n. Since f_n is Riemann integrable, by the integrability criterion (Lesson 3, Theorem 7.6), there exists a partition P with $U(f_n, P) - L(f_n, P) < \varepsilon/3$.

On each subinterval $[x_{i-1}, x_i]$, for any $s, t \in [x_{i-1}, x_i]$:

$$f(s) - f(t) = \big(f(s) - f_n(s)\big) + \big(f_n(s) - f_n(t)\big) + \big(f_n(t) - f(t)\big).$$

Taking the supremum over s and infimum over t:

$$M_i(f) - m_i(f) \le \frac{2\varepsilon}{3(b - a)} + (M_i(f_n) - m_i(f_n)),$$

where $M_i(g) = \sup_{[x_{i-1},x_i]} g$ and $m_i(g) = \inf_{[x_{i-1},x_i]} g$. Therefore:

$$\begin{aligned} U(f,P) - L(f,P) &= \sum_{i=1}^{k} (M_i(f) - m_i(f))(x_i - x_{i-1}) \\ &\le \frac{2\varepsilon}{3(b-a)} \sum_{i=1}^{k} (x_i - x_{i-1}) + \sum_{i=1}^{k} (M_i(f_n) - m_i(f_n))(x_i - x_{i-1}) \\ &= \frac{2\varepsilon}{3} + U(f_n,P) - L(f_n,P) < \frac{2\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon. \end{aligned}$$

By the integrability criterion, f is Riemann integrable.

Step 2: The integrals converge. For $n \ge N$:

$$\begin{aligned} \left| \int_a^b f_n - \int_a^b f \right| = \left| \int_a^b (f_n - f) \right| &\le \int_a^b |f_n - f| \\ &\le (b-a) \sup_{x\in[a,b]} |f_n(x) - f(x)|. \end{aligned}$$

Since $\sup_{x\in[a,b]} |f_n(x) - f(x)| \to 0$, the right side tends to 0, giving $\int_a^b f_n \to \int_a^b f$. □

Example 3.8 (Interchange Fails Under Pointwise Convergence)**.** Define $g_n \colon [0,1] \to \mathbb{R}$ by

$$g_n(x) = \begin{cases} n^2 x & \text{if } 0 \le x \le 1/n, \\ 2n - n^2 x & \text{if } 1/n < x \le 2/n, \\ 0 & \text{if } 2/n < x \le 1. \end{cases}$$

Each g_n is a "tent" function with height n and base $[0, 2/n]$. For every fixed $x > 0$, $g_n(x) = 0$ for all $n > 2/x$, so $g_n \to 0$ pointwise. Also $g_n(0) = 0$ for all n. Hence $g_n \to 0$ pointwise on $[0,1]$. But $\int_0^1 g_n = \frac{1}{2} \cdot \frac{2}{n} \cdot n = 1$ for all n. Thus

$$\lim_{n\to\infty} \int_0^1 g_n = 1 \ne 0 = \int_0^1 \lim_{n\to\infty} g_n.$$

The convergence is not uniform ($\sup |g_n| = n \to \infty$), so Theorem 3.7 does not apply.

Uniform convergence and expected reward computation

In reinforcement learning, the expected cumulative reward under a policy π is $J(\pi) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r_t\right]$. When we approximate the value function V^π by a sequence of iterates V_n, we need the approximation error in the integral $\int V_n \, d\mu$ to vanish. Uniform convergence $\|V_n - V^\pi\|_\infty \to 0$ guarantees $\int V_n \, d\mu \to \int V^\pi \, d\mu$ for any probability measure μ. In Phase 01 (Measure Theory), the dominated convergence theorem provides a more general sufficient condition that relaxes uniform convergence to pointwise convergence plus a dominating integrable function.

We now turn to series of functions, where uniform convergence is established by a powerful comparison test.

4 The Weierstrass M-Test

A *series of functions* $\sum_{n=1}^{\infty} g_n(x)$ converges pointwise (resp. uniformly) on A if the partial sums $S_N(x) = \sum_{n=1}^{N} g_n(x)$ converge pointwise (resp. uniformly) on A.

Theorem 4.1 (Weierstrass M-Test)**.** *Let $\{g_n\}$ be a sequence of functions $g_n\colon A \to \mathbb{R}$. Suppose there exist constants $M_n \geq 0$ such that:*

(i) $|g_n(x)| \leq M_n$ for all $x \in A$ and all $n \geq 1$, and

(ii) $\sum_{n=1}^{\infty} M_n$ converges.

Then $\sum_{n=1}^{\infty} g_n(x)$ converges uniformly (and absolutely) on A.

Proof. Let $S_N(x) = \sum_{n=1}^{N} g_n(x)$. We show $\{S_N\}$ is uniformly Cauchy: for $M > N$,

$$|S_M(x) - S_N(x)| = \left| \sum_{n=N+1}^{M} g_n(x) \right| \leq \sum_{n=N+1}^{M} |g_n(x)| \leq \sum_{n=N+1}^{M} M_n.$$

Since $\sum M_n$ converges, its tail $\sum_{n=N+1}^{\infty} M_n \to 0$ as $N \to \infty$ (Lesson 1, the Cauchy criterion for series). Given $\varepsilon > 0$, choose N_0 such that $\sum_{n=N_0+1}^{\infty} M_n < \varepsilon$. Then for all $M > N \geq N_0$ and *all* $x \in A$:

$$|S_M(x) - S_N(x)| \leq \sum_{n=N+1}^{M} M_n \leq \sum_{n=N_0+1}^{\infty} M_n < \varepsilon.$$

Letting $M \to \infty$, the partial sums converge pointwise to some function $S(x) = \sum_{n=1}^{\infty} g_n(x)$, and the bound above gives $|S(x) - S_N(x)| \leq \varepsilon$ for all $x \in A$ and $N \geq N_0$. This means $\sup_{x \in A} |S(x) - S_N(x)| \leq \varepsilon$ for $N \geq N_0$, i.e., $S_N \rightrightarrows S$.

Absolute convergence follows because $\sum |g_n(x)| \leq \sum M_n < \infty$ for every $x \in A$. □

Example 4.2 (Uniform Convergence of a Power Series)**.** Consider the series $\sum_{n=0}^{\infty} x^n/n!$ on $[-R, R]$ for any fixed $R > 0$. Setting $M_n = R^n/n!$, we have $|x^n/n!| \leq M_n$ for all $|x| \leq R$, and $\sum M_n = e^R < \infty$. By the Weierstrass M-test, the series converges uniformly on $[-R, R]$. Since each partial sum is a polynomial (hence continuous), the uniform limit $e^x = \sum_{n=0}^{\infty} x^n/n!$ is continuous on $[-R, R]$ by Theorem 3.6.

Example 4.3 (The p-Series of Functions)**.** Define $g_n(x) = \sin(nx)/n^2$ on $\mathbb{R}$. Since $|g_n(x)| \leq 1/n^2 = M_n$ and $\sum 1/n^2 = \pi^2/6 < \infty$, the Weierstrass M-test gives uniform convergence of $\sum_{n=1}^{\infty} \sin(nx)/n^2$ on all of $\mathbb{R}$. Each partial sum is continuous, so the limit function is continuous.

Example 4.4 (A Series Where the M-Test Fails)**.** The series $\sum_{n=1}^{\infty} (-1)^{n+1} x^n/n$ converges pointwise on $(-1, 1]$ to $\ln(1 + x)$ (by the alternating series test). However, $\sup_{x \in (-1,1]} |x^n/n| = 1/n$, and $\sum 1/n$ diverges, so the M-test does not apply on the full interval. On any $[-a, a]$ with $0 < a < 1$, we have $|x^n/n| \leq a^n/n$ and $\sum a^n/n < \infty$ (comparison with a geometric series), so uniform convergence holds on $[-a, a]$.

Connection to dominated convergence

The Weierstrass M-test is the series analogue of the dominated convergence theorem from Lebesgue integration theory (Phase 01, Lesson 5). Both results use a "dominating" bound (M_n here, an integrable function g there) to justify interchange of limit and summation or integration. In Phase 01, the dominated convergence theorem will generalize Theorem 3.7 from uniform to pointwise convergence, provided a dominating integrable function exists.

With the theory of function sequences in hand, we now generalize the notion of convergence itself by moving from $\mathbb{R}$ to abstract metric spaces.

5 Metric Spaces

Definition 5.1 (Metric Space)**.** A *metric space* is a pair (X, d) where X is a nonempty set and $d\colon X \times X \to [0, \infty)$ is a function (called a *metric* or *distance function*) satisfying, for all $x, y, z \in X$:

(i) **Identity of indiscernibles:** $d(x, y) = 0$ if and only if $x = y$.
(ii) **Symmetry:** $d(x, y) = d(y, x)$.
(iii) **Triangle inequality:** $d(x, z) \leq d(x, y) + d(y, z)$.

Example 5.2 (Euclidean Metric on $\mathbb{R}^n$)**.** The space $\mathbb{R}^n$ with the Euclidean metric $d(\mathbf{x}, \mathbf{y}) = \left(\sum_{i=1}^n (x_i - y_i)^2\right)^{1/2}$ is a metric space. The triangle inequality is the Cauchy–Schwarz inequality applied to $\mathbb{R}^n$. When $n = 1$, this reduces to $d(x, y) = |x - y|$, the standard metric on $\mathbb{R}$.

Example 5.3 (Discrete Metric)**.** On any nonempty set X, define

$$d(x, y) = \begin{cases} 0 & \text{if } x = y, \\ 1 & \text{if } x \neq y. \end{cases}$$

Verification of the metric axioms is immediate. The triangle inequality holds because $d(x, z) \leq 1 \leq d(x, y) + d(y, z)$ whenever $x \neq z$ (at least one of $d(x, y), d(y, z)$ equals 1). In this metric, every subset is open (since the open ball $B(x, 1/2) = \{x\}$), and every sequence converges if and only if it is eventually constant.

Example 5.4 (Supremum Metric on $C[a, b]$)**.** Let $C[a, b]$ denote the set of continuous functions $f\colon [a, b] \to \mathbb{R}$. Define

$$d_\infty(f, g) = \sup_{x \in [a,b]} |f(x) - g(x)| = \|f - g\|_\infty.$$

This is a metric (the supremum exists by the extreme value theorem, since $|f - g|$ is continuous on the compact set $[a, b]$). Note that $d_\infty(f_n, f) \to 0$ is precisely the statement that $f_n \rightrightarrows f$ on $[a, b]$: uniform convergence *is* convergence in the sup-metric.

Definition 5.5 (Open Ball, Open Set, Closed Set)**.** Let (X, d) be a metric space.
(i) The *open ball* of radius $r > 0$ centered at $x \in X$ is $B(x, r) = \{y \in X : d(x, y) < r\}$.
(ii) A set $U \subseteq X$ is *open* if for every $x \in U$ there exists $r > 0$ such that $B(x, r) \subseteq U$.
(iii) A set $F \subseteq X$ is *closed* if its complement $X \setminus F$ is open.

Proposition 5.6 (Properties of Open and Closed Sets)**.** *In a metric space (X, d):*
(i) $\emptyset$ and X are both open and closed.
(ii) Arbitrary unions of open sets are open.
(iii) Finite intersections of open sets are open.
(iv) A set F is closed if and only if every convergent sequence $\{x_n\} \subseteq F$ has its limit in F.

Proof. (i) $\emptyset$ is open vacuously; X is open since $B(x, 1) \subseteq X$ for all x.

(ii) Let $U = \bigcup_\alpha U_\alpha$ where each U_α is open. If $x \in U$, then $x \in U_\alpha$ for some α, so there exists $r > 0$ with $B(x, r) \subseteq U_\alpha \subseteq U$.

(iii) Let $U = U_1 \cap \cdots \cap U_k$ where each U_i is open. If $x \in U$, then $x \in U_i$ for each i, so there exist $r_i > 0$ with $B(x, r_i) \subseteq U_i$. Setting $r = \min(r_1, \ldots, r_k) > 0$, we get $B(x, r) \subseteq U_i$ for all i, so $B(x, r) \subseteq U$.

(iv) ($\Rightarrow$) Suppose F is closed and $\{x_n\} \subseteq F$ with $x_n \to x$. If $x \notin F$, then $x \in X \setminus F$ which is open, so there exists $r > 0$ with $B(x, r) \subseteq X \setminus F$. But $x_n \to x$ implies $x_n \in B(x, r)$ for large n, contradicting $x_n \in F$.

($\Leftarrow$) Suppose every convergent sequence in F has its limit in F. If $X \setminus F$ is not open, there exists $x \in X \setminus F$ such that for every $n \geq 1$, $B(x, 1/n)$ contains a point $x_n \in F$. Then $d(x_n, x) < 1/n \to 0$, so $x_n \to x$, forcing $x \in F$ by hypothesis—a contradiction. $\square$

Definition 5.7 (Convergence in a Metric Space)**.** A sequence $\{x_n\}$ in a metric space (X, d) *converges* to $x \in X$ if $d(x_n, x) \to 0$ as $n \to \infty$. That is, for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ with $d(x_n, x) < \varepsilon$ for all $n \geq N$. We write $x_n \to x$.

Proposition 5.8 (Uniqueness of Limits)**.** *In a metric space, limits are unique: if $x_n \to x$ and $x_n \to y$, then $x = y$.*

Proof. By the triangle inequality: $d(x, y) \leq d(x, x_n) + d(x_n, y) \to 0 + 0 = 0$. Hence $d(x, y) = 0$, so $x = y$. □

With the basic metric space framework established, we can now define Cauchy sequences and completeness—the properties that make convergence arguments work.

6 Completeness and Compactness

Definition 6.1 (Cauchy Sequence)**.** A sequence $\{x_n\}$ in a metric space (X, d) is a *Cauchy sequence* if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $d(x_m, x_n) < \varepsilon$ for all $m, n \geq N$.

Proposition 6.2. *Every convergent sequence in a metric space is Cauchy.*

Proof. If $x_n \to x$, then for $\varepsilon > 0$, choose N with $d(x_n, x) < \varepsilon/2$ for $n \geq N$. For $m, n \geq N$: $d(x_m, x_n) \leq d(x_m, x) + d(x, x_n) < \varepsilon/2 + \varepsilon/2 = \varepsilon$. □

Definition 6.3 (Complete Metric Space)**.** A metric space (X, d) is *complete* if every Cauchy sequence in X converges to a limit in X.

Example 6.4 ($\mathbb{R}$ Is Complete; $\mathbb{Q}$ Is Not)**.** The real line $(\mathbb{R}, |\cdot|)$ is complete—this is the content of the Cauchy convergence criterion (Lesson 1, Theorem 4.10). In contrast, $\mathbb{Q}$ is not complete: the sequence $x_1 = 1, x_{n+1} = \frac{1}{2}(x_n + 2/x_n)$ is Cauchy in $\mathbb{Q}$ (Newton's method applied to $f(x) = x^2 - 2$), but its limit $\sqrt{2}$ is not in $\mathbb{Q}$.

Theorem 6.5 ($C[a, b]$ with the Sup-Metric Is Complete)**.** *The metric space $(C[a, b], d_\infty)$ is complete.*

Proof. Let $\{f_n\}$ be a Cauchy sequence in $(C[a, b], d_\infty)$. We must show it converges to some $f \in C[a, b]$.

Step 1: Pointwise limit exists. For each fixed $x \in [a, b]$, $|f_m(x) - f_n(x)| \leq d_\infty(f_m, f_n)$, so $\{f_n(x)\}$ is a Cauchy sequence in $\mathbb{R}$. Since $\mathbb{R}$ is complete, $f_n(x)$ converges to some value, which we call $f(x)$. This defines a function $f\colon [a, b] \to \mathbb{R}$.

Step 2: Convergence is uniform. Given $\varepsilon > 0$, choose N such that $d_\infty(f_m, f_n) < \varepsilon$ for all $m, n \geq N$. For any fixed $x \in [a, b]$ and $n \geq N$, take $m \to \infty$ in $|f_m(x) - f_n(x)| \leq \varepsilon$ (using continuity of $|\cdot|$) to get $|f(x) - f_n(x)| \leq \varepsilon$. Since this holds for every x, $\sup_{x \in [a,b]} |f(x) - f_n(x)| \leq \varepsilon$ for all $n \geq N$. Thus $f_n \rightrightarrows f$.

Step 3: f is continuous. Each f_n is continuous and $f_n \rightrightarrows f$, so f is continuous by Theorem 3.6. Hence $f \in C[a, b]$ and $d_\infty(f_n, f) \to 0$. □

Completeness of function spaces in RL

The completeness of $(C[a, b], d_\infty)$ is a prototype for a central pattern in reinforcement learning theory. The space of bounded functions $B(S)$ on a state space S, equipped with the sup-norm $\|V\|_\infty = \sup_{s \in S} |V(s)|$, is also complete (by the same argument: pointwise limits of Cauchy sequences exist in $\mathbb{R}$, and the convergence is uniform). This is the space in which value functions live, and its completeness is what makes the contraction mapping theorem applicable. In Phase 01 (Functional Analysis), we will study Banach spaces—complete normed vector spaces—as the general framework for these arguments.

Definition 6.6 (Sequential Compactness)**.** A metric space (X, d) is *sequentially compact* if every sequence in X has a convergent subsequence (with limit in X).

Theorem 6.7 (Heine–Borel for $\mathbb{R}^n$)**.** *A subset $K \subseteq \mathbb{R}^n$ (with the Euclidean metric) is sequentially compact if and only if it is closed and bounded.*

Proof. ($\Leftarrow$) Let $K \subseteq \mathbb{R}^n$ be closed and bounded, and let $\{x_k\} \subseteq K$. Since K is bounded, $K \subseteq [-M, M]^n$ for some $M > 0$. Write $x_k = (x_k^{(1)}, \ldots, x_k^{(n)})$. The sequence $\{x_k^{(1)}\}$ is bounded, so by Bolzano–Weierstrass (Lesson 1, Theorem 4.7), it has a convergent subsequence. Passing to this subsequence, the second coordinates $\{x_k^{(2)}\}$ are bounded, so they also have a convergent subsequence. Repeating n times (each time passing to a further subsequence), we obtain a subsequence $\{x_{k_j}\}$ such that every coordinate converges. Hence $x_{k_j} \to x$ for some $x \in \mathbb{R}^n$. Since K is closed, $x \in K$.

($\Rightarrow$) If K is not bounded, choose $x_k \in K$ with $\|x_k\| \geq k$; no subsequence can converge. If K is not closed, choose $x_k \in K$ with $x_k \to x \notin K$; every convergent subsequence must also converge to $x \notin K$, contradicting sequential compactness. □

Example 6.8. The interval $[0, 1] \subset \mathbb{R}$ is sequentially compact (closed and bounded). The open interval $(0, 1)$ is bounded but not closed: the sequence $1/n \to 0 \notin (0, 1)$ shows it is not sequentially compact. The set $\mathbb{R}$ itself is closed but not bounded, hence not sequentially compact: the sequence n has no convergent subsequence.

We now have all the ingredients for the main result of this lesson.

7 The Contraction Mapping Theorem

Definition 7.1 (Contraction Mapping)**.** Let (X, d) be a metric space. A function $T\colon X \to X$ is a *contraction mapping* (or *contraction*) if there exists a constant $\gamma \in [0, 1)$ such that

$$d(T(x), T(y)) \leq \gamma\, d(x, y) \quad \text{for all } x, y \in X.$$

The constant γ is called the *contraction factor*.

Remark 7.2. Every contraction is (uniformly) continuous: given $\varepsilon > 0$, taking $\delta = \varepsilon$ gives $d(T(x), T(y)) \leq \gamma\, d(x, y) < d(x, y) < \varepsilon$ whenever $d(x, y) < \delta$. In fact, every contraction is Lipschitz continuous with Lipschitz constant $\gamma < 1$.

Banach Fixed-Point Theorem

Theorem 7.3 (Banach Fixed-Point Theorem (Contraction Mapping Theorem))**.** *Let (X, d) be a complete metric space and let $T\colon X \to X$ be a contraction with contraction factor $\gamma \in [0, 1)$. Then:*

(i) ***Existence.*** *T has a fixed point: there exists $x^* \in X$ with $T(x^*) = x^*$.*

(ii) ***Uniqueness.*** *The fixed point is unique.*

(iii) ***Convergence rate.*** *For any starting point $x_0 \in X$, the iterates $x_n = T(x_{n-1}) = T^n(x_0)$ satisfy*

$$d(x_n, x^*) \leq \frac{\gamma^n}{1 - \gamma}\, d(x_0, x_1).$$

Proof. Fix any $x_0 \in X$ and define $x_n = T(x_{n-1})$ for $n \geq 1$.

Step 1: $\{x_n\}$ is Cauchy. First, observe that

$$d(x_{n+1}, x_n) = d(T(x_n), T(x_{n-1})) \leq \gamma\, d(x_n, x_{n-1}) \leq \cdots \leq \gamma^n\, d(x_1, x_0).$$

For $m > n$, the triangle inequality gives:

$$\begin{aligned} d(x_m, x_n) &\le \sum_{k=n}^{m-1} d(x_{k+1}, x_k) \le \sum_{k=n}^{m-1} \gamma^k \, d(x_1, x_0) \\ &= \gamma^n \sum_{j=0}^{m-n-1} \gamma^j \, d(x_1, x_0) \le \gamma^n \sum_{j=0}^{\infty} \gamma^j \, d(x_1, x_0) = \frac{\gamma^n}{1-\gamma} \, d(x_1, x_0). \end{aligned}$$

Since $\gamma \in [0, 1)$, $\gamma^n \to 0$, so the right side tends to 0 as $n \to \infty$. Hence $\{x_n\}$ is Cauchy.

Step 2: Existence. Since (X, d) is complete, $x_n \to x^*$ for some $x^* \in X$. We show $T(x^*) = x^*$. Since T is continuous (Remark 7.2):

$$T(x^*) = T\left(\lim_{n\to\infty} x_n\right) = \lim_{n\to\infty} T(x_n) = \lim_{n\to\infty} x_{n+1} = x^*.$$

Step 3: Uniqueness. Suppose $T(y^*) = y^*$ for some $y^* \in X$. Then

$$d(x^*, y^*) = d(T(x^*), T(y^*)) \le \gamma \, d(x^*, y^*).$$

Since $\gamma < 1$, this implies $(1-\gamma) \, d(x^*, y^*) \le 0$, so $d(x^*, y^*) = 0$, hence $x^* = y^*$.

Step 4: Convergence rate. From Step 1, for $m > n$: $d(x_m, x_n) \le \frac{\gamma^n}{1-\gamma} \, d(x_1, x_0)$. Taking $m \to \infty$ and using continuity of the metric:

$$d(x^*, x_n) = \lim_{m\to\infty} d(x_m, x_n) \le \frac{\gamma^n}{1-\gamma} \, d(x_1, x_0) = \frac{\gamma^n}{1-\gamma} \, d(x_0, x_1). \qquad \square$$

Example 7.4 (Contraction on $\mathbb{R}$)**.** Let $T\colon [0, \infty) \to [0, \infty)$ be defined by $T(x) = \frac{1}{2}(x + 2/x)$ (Newton's method for $\sqrt{2}$). For $x, y \ge 1$:

$$|T(x) - T(y)| = \left| \frac{1}{2}(x - y) - \frac{(x-y)}{xy} \right| = \frac{|x-y|}{2} \left| 1 - \frac{2}{xy} \right|.$$

On $[1, 2]$, $xy \ge 1$ and $2/(xy) \le 2$, so this analysis requires care. Instead, note that T maps $[\sqrt{2}, 2]$ into itself (since $T(x) \ge \sqrt{2}$ by AM-GM and $T(x) \le x \le 2$ for $x \ge \sqrt{2}$), and on $[\sqrt{2}, 2]$, $|T'(x)| = \frac{1}{2}|1 - 2/x^2| \le 1/2$ by the MVT (Lesson 3, Theorem 5.4). Hence T is a contraction with $\gamma = 1/2$ on $[\sqrt{2}, 2]$, and the unique fixed point is $x^* = \sqrt{2}$. Starting from $x_0 = 2$: $x_1 = 3/2 = 1.5$, $x_2 = 17/12 \approx 1.4167$, converging rapidly to $\sqrt{2} \approx 1.4142$.

Example 7.5 (Non-Example: $\gamma = 1$ Fails)**.** The map $T\colon [0, \infty) \to [0, \infty)$ defined by $T(x) = x+1$ satisfies $|T(x) - T(y)| = |x - y|$ (so $\gamma = 1$, not strictly less than 1). It has no fixed point: $T(x) = x$ would require $x + 1 = x$.

Similarly, the map $T(x) = x/(1 + x)$ on $[0, \infty)$ satisfies $|T(x) - T(y)| \le |x - y|$ (with γ approaching 1 as $x, y \to 0$). It has a unique fixed point $x^* = 0$, but the convergence rate is only $O(1/n)$, not geometric. The strict inequality $\gamma < 1$ in the Banach theorem is essential for the geometric rate.

Example 7.6 (Value Iteration as a Contraction)**.** Consider a finite Markov decision process with state space $S = \{1, \ldots, |S|\}$, action space A, rewards bounded by $R_{\max}$, transition probabilities $p(s'|s, a)$, and discount factor $\gamma \in (0, 1)$. The Bellman optimality operator $T\colon \mathbb{R}^{|S|} \to \mathbb{R}^{|S|}$ is defined by

$$(TV)(s) = \max_{a\in A} \left[r(s, a) + \gamma \sum_{s'\in S} p(s'|s, a) \, V(s') \right].$$

For any $V, W \in \mathbb{R}^{|S|}$ and any state s:

$$\begin{aligned}|(TV)(s) - (TW)(s)| &= \left|\max_a \left[r(s,a) + \gamma \sum_{s'} p(s'|s,a)V(s')\right] - \max_a \left[r(s,a) + \gamma \sum_{s'} p(s'|s,a)W(s')\right]\right| \\ &\le \max_a \gamma \sum_{s'} p(s'|s,a)\,|V(s') - W(s')| \\ &\le \gamma \|V - W\|_\infty,\end{aligned}$$

where the first inequality uses $|\max_a f(a) - \max_a g(a)| \le \max_a |f(a) - g(a)|$. Taking the supremum over s: $\|TV - TW\|_\infty \le \gamma \|V - W\|_\infty$.

Since $(\mathbb{R}^{|S|}, \|\cdot\|_\infty)$ is a complete metric space and T is a γ-contraction, the Banach fixed-point theorem gives:

(i) There exists a unique $V^* \in \mathbb{R}^{|S|}$ with $TV^* = V^*$ (the Bellman optimality equation).

(ii) For any initial V_0, the value iteration sequence $V_n = TV_{n-1}$ satisfies $\|V_n - V^*\|_\infty \le \frac{\gamma^n}{1-\gamma}\|V_0 - V_1\|_\infty$.

With $\gamma = 0.99$, reaching ε-accuracy requires about $n \ge \frac{\log(\varepsilon(1-\gamma)/\|V_0 - V_1\|_\infty)}{\log \gamma}$ iterations.

8 Exercises

Exercise 8.1 (Sup-norm characterization of uniform convergence). $(*)$
Let $f_n, f\colon [a,b] \to \mathbb{R}$ be bounded functions. Prove that $f_n \rightrightarrows f$ on $[a,b]$ if and only if $\|f_n - f\|_\infty \to 0$, where $\|g\|_\infty = \sup_{x\in[a,b]} |g(x)|$.

Exercise 8.2 (Weierstrass M-test application). $(*)$
Show that the series $\sum_{n=1}^\infty \frac{\cos(nx)}{n^3}$ converges uniformly on $\mathbb{R}$. Conclude that its sum is a continuous function.

Exercise 8.3 (Completeness of ℓ^∞). $(**)$
Let ℓ^∞ be the set of bounded real sequences $\mathbf{x} = (x_1, x_2, \ldots)$ with the metric $d(\mathbf{x}, \mathbf{y}) = \sup_{n\ge1} |x_n - y_n|$. Prove that (ℓ^∞, d) is a complete metric space.

Hint: Mimic the proof of Theorem 6.5: show that a Cauchy sequence in ℓ^∞ converges coordinatewise, the convergence is uniform, and the limit is bounded.

Exercise 8.4 (Failure of the contraction mapping theorem without completeness). $(**)$
Let $X = (0,1]$ with the standard metric $d(x,y) = |x - y|$ and define $T\colon X \to X$ by $T(x) = x/2$.

(a) Show that T is a contraction with factor $\gamma = 1/2$.

(b) Show that T has no fixed point in X.

(c) Explain why this does not contradict the Banach fixed-point theorem.

Exercise 8.5 (Value iteration convergence rate). $(**)$
Consider a two-state MDP with $S = \{s_1, s_2\}$, a single action, rewards $r(s_1) = 1$, $r(s_2) = 0$, transition probabilities $p(s_1|s_1) = p(s_2|s_2) = 1$ (absorbing states), and discount factor $\gamma = 0.9$.

(a) Write the Bellman operator T explicitly and verify that $V^* = (10, 0)$.

(b) Starting from $V_0 = (0,0)$, compute V_1, V_2, V_3 and verify the bound $\|V_n - V^*\|_\infty \le \frac{\gamma^n}{1-\gamma}\|V_0 - V_1\|_\infty$.

(c) How many iterations are needed to guarantee $\|V_n - V^*\|_\infty < 0.01$?

Exercise 8.6 (Uniform convergence and differentiation). $(***)$
The interchange of limit and derivative is more delicate than the interchange of limit and integral. Consider $f_n(x) = \frac{\sin(nx)}{n}$ on $\mathbb{R}$.

(a) Show that $f_n \rightrightarrows 0$ on $\mathbb{R}$.

(b) Compute $f_n'(x)$ and show that $\{f_n'\}$ does not converge pointwise at $x = 0$.

(c) Conclude that uniform convergence of $\{f_n\}$ does not imply convergence of $\{f_n'\}$. Compare with Theorem 3.7: why is differentiation harder than integration?

Hint: For (c), note that differentiation amplifies oscillation (multiplies by frequency n), while integration averages it out.

9 Summary

- **Pointwise convergence** $f_n \to f$ requires $|f_n(x) - f(x)| < \varepsilon$ for each x separately, with N depending on both x and ε; it can destroy continuity of the limit (e.g., x^n on $[0, 1]$).
- **Uniform convergence** $f_n \rightrightarrows f$ requires a single $N(\varepsilon)$ that works for all x simultaneously, equivalently $\|f_n - f\|_\infty \to 0$; it preserves continuity and permits interchange of limit and Riemann integral.
- The **Weierstrass M-test** establishes uniform convergence of a series $\sum g_n$ by bounding $|g_n(x)| \leq M_n$ with a convergent numerical series $\sum M_n$. It is the function-series analogue of the dominated convergence theorem.
- A **metric space** (X, d) abstracts the notion of distance, enabling convergence, continuity, and completeness to be studied in general settings (function spaces, sequence spaces, etc.).
- **Completeness** (every Cauchy sequence converges) is the key property that distinguishes $\mathbb{R}$ from $\mathbb{Q}$ and makes fixed-point arguments work. The function space $(C[a, b], \|\cdot\|_\infty)$ is complete.
- **Sequential compactness** generalizes the Bolzano–Weierstrass theorem; in $\mathbb{R}^n$ (Heine–Borel), a set is sequentially compact if and only if it is closed and bounded.
- The **Banach fixed-point theorem** states that a contraction T on a complete metric space has a unique fixed point x^*, and the iterates $T^n(x_0)$ converge to x^* at a geometric rate $O(\gamma^n)$.
- The Bellman optimality operator is a γ-contraction on the complete space of bounded value functions, so **value iteration converges** to the unique optimal value function at rate $O(\gamma^n)$.
- Pointwise convergence can fail to commute with integration (tent function example) or differentiation ($\sin(nx)/n$); uniform convergence fixes the integration issue but not the differentiation one.

As the capstone lesson of the Real Analysis subdomain, we summarize the full arc of Lessons 1–4 in the following table.

10 Further Reading

- In **Phase 01** (Functional Analysis, Lessons 6–7), the metric space framework is extended to *normed spaces* and *Banach spaces* (complete normed spaces), providing the natural setting for the contraction mapping theorem in infinite dimensions. *Hilbert spaces* add inner product structure, enabling orthogonal projections and the Riesz representation theorem.
- In **Phase 01** (Measure Theory, Lessons 4–5), the interchange of limit and integral is treated definitively via the *monotone convergence theorem* and the *dominated convergence theorem*, which generalize Theorem 3.7 by replacing uniform convergence with weaker hypotheses.

Table 1: Real Analysis Subdomain: Lessons 1–4 Overview

Lesson	Core Topics	Key RL/ML Applications
1	Real numbers, sequences, series (completeness, Cauchy criterion, Bolzano–Weierstrass)	Convergence of iterative algorithms, existence of suprema/infima for optimization
2	Topology, continuity, compactness on $\mathbb{R}$ (EVT, IVT, uniform continuity)	Continuous dependence of policies on parameters, extreme value theorem for bounded rewards
3	Differentiation, integration (chain rule, MVT, Taylor, Riemann integral, FTC)	Gradient descent (MVT, Taylor), backpropagation (chain rule), expected reward computation
4	Function sequences, metric spaces (uniform convergence, completeness, Banach fixed-point theorem)	Value iteration convergence (contraction mapping theorem), function space analysis

- In **Phase 02** (Core RL, Lessons 13–14), the Banach fixed-point theorem is applied to the Bellman operators for both policy evaluation and optimal control, yielding convergence of value iteration and policy iteration with explicit rate bounds.
- Rudin [1976, Chapter 7] continues with *equicontinuity* and the *Arzelà–Ascoli theorem*, characterizing compact subsets of $C[a, b]$—essential for proving existence of optimal policies in continuous-state MDPs.
- Pugh [2015, Chapters 2–3] develops metric space topology in greater depth, including the Baire category theorem and its applications to analysis.
- The Stone–Weierstrass theorem (Rudin [1976, Theorem 7.26]) shows that polynomials are dense in $C[a, b]$ under the sup-norm, providing a classical density result that foreshadows neural network universal approximation theorems (Phase 03).
- For historical context, Banach's original 1922 paper introduced the fixed-point theorem in the setting of complete metric spaces. The theorem was independently discovered by Caccioppoli in 1931. Bartle and Sherbert [2011, Chapter 9] provides a textbook treatment with many applications.

References

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Walter Rudin. *Principles of Mathematical Analysis*. McGraw-Hill, 3rd edition, 1976.